Lab Session 8

OBJECTIVE(S)

- Learn about Relational Database Management Systems
- Learn about Creating Database in Microsoft Access
- Learn about Creating Relationships
- Learn about Creating Queries

What is a Database?

A database is a collection of organized data that can easily be accessed, managed, and updated.

Database Management Systems

A Database Management System (DBMS) is a software program that enables the creation and management of databases. Generally, these databases will be more complex than the text files/spreadsheets.

Relational Database Management System

A relational Database Management System is a type of database management system (DBMS) which stores data in the form of related tables.

Popular RDBMS

- MS Access
- Oracle
- MySQL
- Microsoft SQL Server

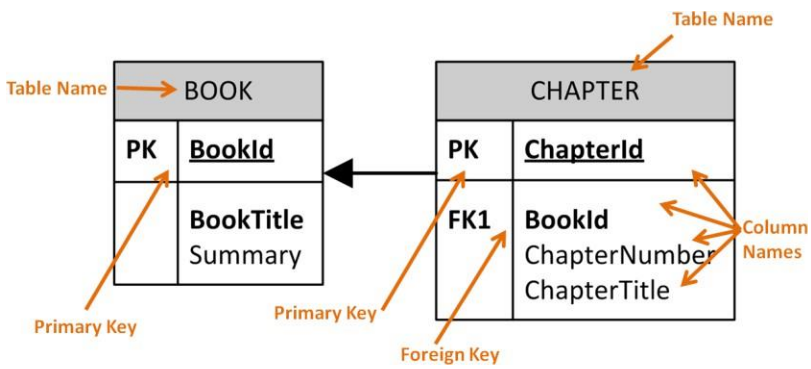
Table

Customer Table

CustomerID	CustomerName	Address	City	Country
1	Alan Turing	13 Guild Street	London	UK
2	Dennis Ritchie	61 Farnum Road	New York	USA
3	John Neumann	Csabai kapu 4	Budapest	USA
4	Ada Lovelace	68 Crown Street	London	UK

Database Relationship

Book has a Chapter:



<u>Columns</u>

A field also called a column in a relational database, is part of a table that is assigned a specific data type. The data type determines what kind of data the column is allowed to hold. This enables the designer of the table to help maintain the integrity of the data.

Every database table must consist of at least one column. Columns are those elements within a table that hold specific types of data, such as a person's name or address. For example, a valid column in a customer table might be the customer's name.

<u>Rows</u>

A row is a record of data in a database table. For example, a row of data in a customer table might consist of a particular customer's identification number, name, address, phone number, and fax number. A row is composed of fields that contain data from one record in a table. A table can contain as little as one row of data and up to as many as millions of rows of data or records.

<u>Integrity Constraints</u>

Integrity constraints ensure the accuracy and consistency of data in a relational database. Data integrity is handled in a relational database through the concept of referential integrity. Many types of integrity constraints play a role in referential integrity (RI).

Referential integrity is a data quality concept that ensures that when you make changes to data in one place, those changes are reflected in other related records

<u>Primary Key</u>

The primary key of a relational table uniquely identifies each record in the table. It can either be a normal attribute that is guaranteed to be or it can be generated by the DBMS. Primary keys may consist of a single attribute or multiple attributes in combination.

Example

Imagine we have a Students table that contains a record for each student at a university. The student's unique student ID number would be a good choice for a primary key in the STUDENTS table. The student's first and last name would not be a good choice, as there is always the chance that more than one student might have the same name.

Foreign Key

A foreign key is a key used to link two tables together. A FOREIGN KEY in one table points to a PRIMARY KEY in another table. Let's illustrate the foreign key with an example. Look at the following two tables:

Person Table

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

Orders Table

O_Id	OrderNo	P_Id
1	77895	3
2	44678	3
3	22456	2

- The "P_Id" column in the Orders table points to the "P_Id" column in the Person table.
- The "P_Id" column in the Person table is the PRIMARY KEY in the Person table.
- The P_Id column in the Orders table is a FOREIGN KEY in the Orders table.
- The FOREIGN KEY constraint is used to prevent actions that would destroy links between tables. It also prevents invalid data from being inserted into the foreign key column, because it has to be one of the values contained in the table it points to.

FAMILIARIZATION WITH SOME BASIC DATABASE RELATED TERMS

USING MICROSOFT ACCESS

Microsoft Access is a Database Management System (DBMS) from Microsoft that combines the relational Microsoft Jet Database Engine with a graphical user interface and software development tools. It is a member of the Microsoft Office suite of applications, included in the professional and higher editions

- A ***record*** in a row on a datasheet and is a set of values defined by fields. In a mailing list table, each record would contain the data for one person as specified by the intersecting fields.

- A ***field*** is a column on a datasheet and defines a data type for a set of values in a table. For a mailing list table might include fields for first name, last name, address, city, state, zip code, and telephone number.

- A ***table*** is a grouping of related data organized in fields (columns) and records (rows) on a datasheet. By using a common field in two tables, the data can be combined. Many tables can be stored in a single database.

- A ***database*** is a collection of related information.

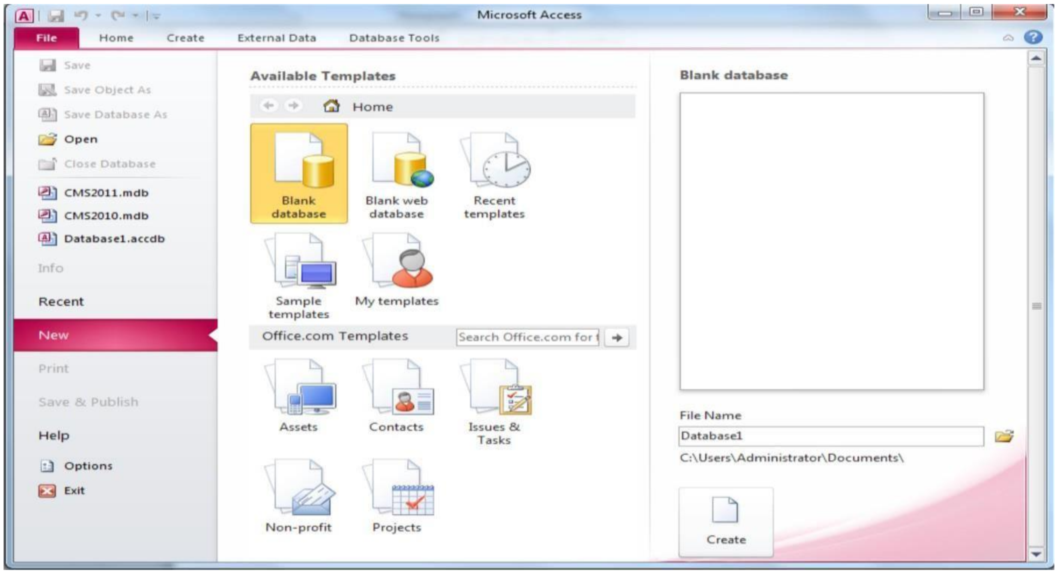


Figure 11.1 MS Access Welcome window

GETTING STARTED

After opening Access, you will be presented with two major options as shown in figure above. First if you are creating a new database or the second if you want to open and edit an existing database. Details about both of these choices are given below:

Open an existing database

If the database was opened recently on the computer, it will be listed in the File Menu either directly or inside *Recent* option. Highlight the database name and click *OK.* Otherwise, click Open (Ctrl + O) from File Menu, navigate to proper folder location, highlight the database name in the listing and click *OK.*

Creating a new database

Click *New* (Ctrl + N) option inside the File Menu. You will be presented with MS Access available database templates both offline and online and then you are required to select any one of them to create your new database. Some of the example offline database templates include Blank Database, Blank Web database, sample database etc. Some of the example online database templates include Assets, Contacts, Issues and Tasks etc. Unlike Word documents, Excel worksheets, and Power Point presentations, you must save an Access database before you start working on it. After selecting "Blank Access database" for example, you will first be prompted to specify a location and name for the database.

<u>Creating Database in Microsoft Access</u>

With database management systems, you need to create your tables before you can enter data. Microsoft Access makes creating tables extremely easy. In fact, when you create a database, Access creates your first table for you.

Using our blank database, we are going to rename Table1 to Customers. This table will have 4 columns: CustomerId, FirstName, LastName, and DateCreated.

Steps

1. Rename the ID field to CustomerId. To do this, Right-click on the ID column and select Rename Field. Enter CustomerId when prompted.
2. In the next field, click on Click to Add and select Short Text.
3. At this point, Access will conveniently highlight/select the field name (currently Field1) so that you can name the field. Call it FirstName.
4. Do the same again for the next field (LastName) and select data type Short Text.
5. In the next field, select the Date & Time data type and name the field DateCreated.

Database Window

The Database Window (figure 11.2) organizes all of the objects in the database e.g. Tables, Queries, Forms etc. The default table listing provides links for creating or modifying tables and will list all of the tables in the database when they have been added.

Design View

Design View (Figure 11.3) customizes the fields in the database so that data can be entered.

Datasheet View

The datasheet view (Figure 11.4) allows you to enter data into the database.

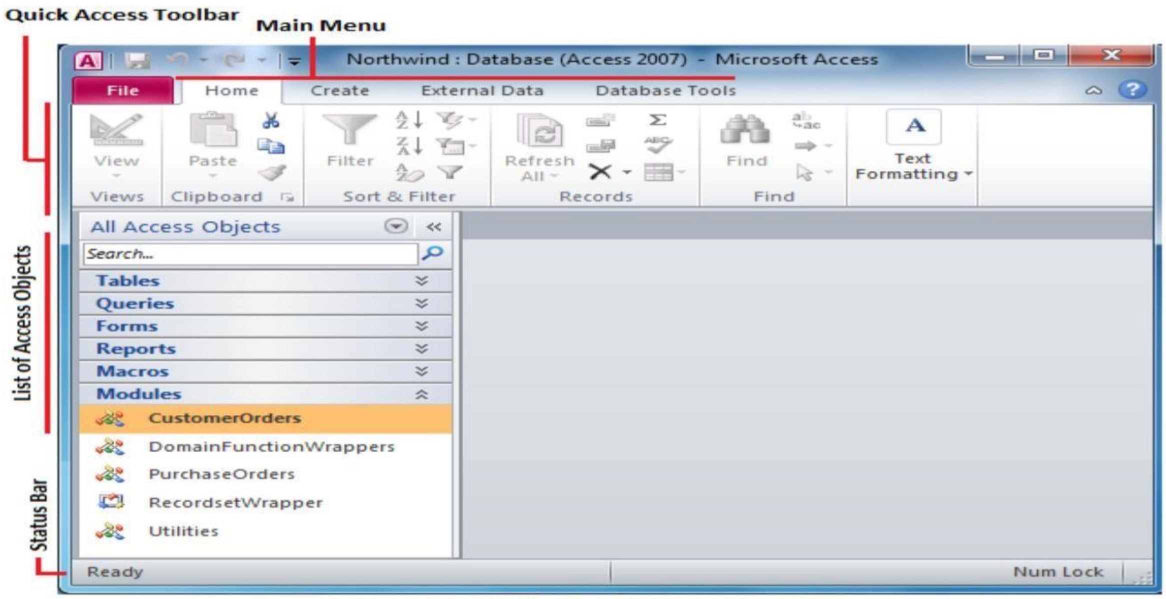


Figure 11.2: Database Window

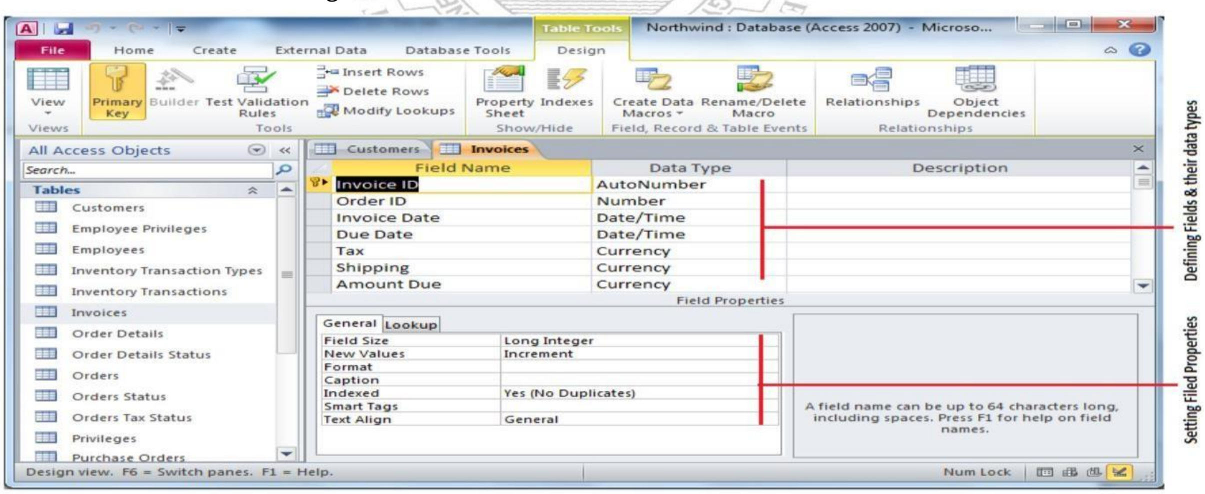


Figure 11.3: Design View

<u>Design view</u>

Design view enables you to design and set up your database. It is kind of a "Behind the scenes" view of your database. This is where you set up and configure your tables, forms, reports, etc

Switch to Design View: The following steps demonstrate how to switch to the Design view.

Steps

1. Click on the View icon at the top left (just under the File menu).
2. At this point, Access will ask you to name and save the table you just created. Type Customers and click OK.
3. You are now looking at your Customer table in Design view. Click on the Date Created field, then in the bottom pane, enter =Now() in the Default Value field. Also, click in the Format field and select General Date from the contextual menu.

CREATING TABLES

Introduction to Tables

Tables are grids that store information in a database similar to the way an Excel worksheet stores information in a workbook. To create a table Click Create from Main Menu. Access provides two ways to create a table for which there are icons Table and Table design. Double-click on any icon to create a table.

- ***Create table in Design view*** will allow you to create the fields of the table. This is the most common way of creating a table and is explained in detail below.
- ***Create table by entering data*** will give you a blank datasheet with just ID field and the rest with unlabeled columns that looks much like an Excel worksheet. Enter data into the cells and click the *Save* button. After the table is saved, the empty cells of the datasheet are trimmed. The fields are given generic names such as "Field1", "Field2", etc. To rename them with more descriptive titles that reflect the content of the fields, change the view to Design view again and modify names as per requirements.

Create a Table in Design View

Design View will allow you to define the fields in the table before adding any data to the datasheet. The window is divided into two parts: a top pane for entering the field name, data type, and an option description of the field, and a bottom pane for specifying field properties as shown in (figure 11.3)

Field Name - This is the name of the field and should represent the contents of the field such as "Name", "Address", "Final Grade", etc. The name cannot exceed 64 characters in length and may include spaces.

Data Type is the type of value that will be entered into the fields.

Text - The default type, text type allows any combination of letters and numbers up to a maximum of 255 characters per field record.

Memo - A text type that stores up to 64,000 characters. oNumber - Any number can be stored.

Date/Time - A date, time, or combination of both.

Currency - Monetary values that can be set up to automatically include a dollar sign ($) and correct decimal and comma positions.

AutoNumber - When a new record is created, Access will automatically assign a unique integer to the record in this field. From the General options, select Increment if the numbers should be assigned in order or random if any random number should be chosen. Since every record in a datasheet must include at least one field that distinguishes it from all others, this is a useful data type to use if the existing data will not produce such values.

Yes/No - Use this option for True/False, Yes/No, On/Off, or other values that must be only one of two.

OLE Object - An OLE (Object Linking and Embedding) object is a sound, picture, or other object such as a Word document or Excel spreadsheet that is created in another program. Use this data type to embed an OLE object or link to the object in the database.

Hyperlink - A hyperlink will link to an Internet or Intranet site, or another location in the database. The data consists of up to four parts each separated by the pound sign (#): DisplayText#Address#SubAddress#ScreenTip. The Address is the only required part of the string. Examples:

Internet hyperlink example:	FGCU Home Page#http://www.fgcu.edu#
Database link example:	#c:\My Documents\database.mdb#MyTable

- **Description** (optional) - Enter a brief description of what the contents of the field are.
- **Field Properties** - Select any pertinent properties for the field from the bottom pane.

Field Properties

Properties for each field are set from the bottom pane of the Design View window.

▪ **Field Size** is used to set the number of characters needed in a text or number field. The default field size for the text type is 50 characters. If the records in the field will only have two or three characters, you can change the size of the field to save disk space or prevent entry errors by limiting the number of characters allowed. Likewise, if the field will require more than 50 characters, enter a number up to 255. The field size is set in exact characters for Text type, but options are given for numbers:

- **Byte** - Positive integers between 1 and 255
- **Integer** - Positive and negative integers between -32,7613 and 32,7613
- **Long Integer (default)** - Larger positive and negative integers between -2 billion and 2 billion.
- **Single** - Single-precision floating-point number
- **Double** - Double-precision floating-point number
- **Decimal** - Allows for Precision and Scale property control

▪ **Format** confirms the data in the field to the same format when it is entered into the datasheet. For text and memo fields, this property has two parts that are separated by a semicolon. The first part of the property is used to apply to the field and the second applies to empty fields.

Primary Key

Every record in a table must have a primary key that differentiates it from every other record in the table. In some cases, it is only necessary to designate an existing field as the primary key if you are certain that every record in the table will have a different value for that particular field. A social security number is an example of a record whose values will only appear once in a database table.

Figure 11.5 Quick Access Toolbar – Table Tools

Designate the primary key field by right-clicking on the record and selection *Primary Key* from the shortcut menu or select *Primary Key* from the Quick Access Toolbar as given in figure above. The primary key field will be noted with a key image to the left. To remove a primary key, repeat one of these steps.

Indexes

Creating indexes allows Access to query and sort records faster. To set an indexed field, select a field that is commonly searched and change the Indexed property to *Yes (Duplicates OK)* if multiple entries of the same data value are allowed or *Yes (No Duplicates)* to prevent duplicates.

Field Validation Rules

Validation Rules specify requirements (change word) for the data entered in the worksheet. A customized message can be displayed to the user when data that violates the rule setting is entered. Click the expression builder ("**...**") button at the end of the Validation Rule box to write the validation rule. Examples of field validation rules include <> *0* to not allow zero values in the record, and *???* to only all data strings three characters in length.

Input Masks

An input mask controls the value of a record and sets it in a specific format. They are similar to the Format property, but instead display the format on the datasheet before the data is entered. For example, a telephone number field can formatted with an input mask to accept ten digits that are automatically formatted as "(555) 123-4567". The blank field would look like (___) ___-____. Add an input mask to a field by following these steps:

- In design view, place the cursor in the field that the input mask will be applied to.
- Click in the white space following *Input Mask* under the *General* tab.
- Click the "**...**" button to use the wizard or enter the mask, (@@@) @@@-@@@@, into the field provided.

DATASHEET RECORDS

Adding Records

Add new records to the table in datasheet view by typing in the record beside the asterisk (*) that marks the new record.

Editing Records

To edit records, simply place the cursor in the record that is to be edited and make the necessary changes. Use the arrow keys to move through the record grid. The previous, next, first, and last record buttons at the bottom of the datasheet are helpful in maneuvering through the datasheet.

Deleting Records

Delete a record on a datasheet by placing the cursor in any field of the record row and select *Delete Record* from the menu bar or click the *Delete* button on the Quick Access toolbar.

Adding and Deleting Columns

Although it is best to add new fields (displayed as columns in the datasheet) in design view because more options are available, they can also be quickly added in datasheet view. Highlight the column that the new column should appear to the left of by clicking its label at the top of the datasheet and press right click then select *Insert field* from the dropdown menu. Entire columns can be deleted by placing the cursor in the column and selecting *Delete field* from the dropdown menu.

Freezing and unfreezing Columns

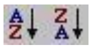
Similar to freezing panes in Excel, columns on an Access table can be frozen. This is helpful if the datasheet has many columns and relevant data would otherwise not appear on the screen at the same time. Freeze a column by placing the cursor in any record in the column and select *Freeze fields* from the right click dropdown men. Select the same option to unfreeze a single column or select *Unfreeze All fields.*

Finding Data in a Table

Open the table in datasheet view. Place the cursor in any record in the field that you want to search and select *Home|Find (Ctrl + F)...* from the Quick Access toolbar. Enter the value criteria in the *Find What:* box. From the *Look In:* drop-down menu, define the area of the search by selecting the entire table or just the field in the table you placed your cursor in during step 2. Select the matching criteria from *Match:* to and click the *More* >> button for additional search parameters. When all of the search criteria are set, click the *Find Next* button. If more than one record meets the criteria, keep clicking *Find Next* until you reach the correct record.

Sorting & Filtering

Sorting and filtering allow you to view records in a table in a different way either by reordering all of the records in the table or view only those records in a table that meet certain criteria that you specify.

Sorting 

In table datasheet view, place the cursor in the column that you want to sort by. Select *Ascending* or *Descending* from the Quick Access toolbar under Home main menu or click the *Sort A to Z* or *Sort Z to A* from the dropdown menu.

To sort by more than one column (such as sorting by date and then sorting records with the same date alphabetically), highlight the columns by clicking and dragging the mouse over the field labels and select one of the sort methods stated above.

Filter by Selection 

This feature will filter records that contain identical data values in a given field such as filtering out all of the records that have the value "Smith" in a name field. To Filter by Selection, place the cursor in the field that you want to filter the other records by and click the *Selection* button inside Home main menu on the toolbar or directly select the criteria from right click dropdown menu.

Filter by Form 

If the table is large, it may be difficult to find the record that contains the value you would like to filter by so using Filter by Form may be advantageous instead. This method creates a blank version of the table with drop-down menus for each field that each contains the values found in the records of that field.

The following methods can be used to select records based on the record selected by that do not have exactly the same value. Type these formats into the field where the drop-down menu appears instead of selecting an absolute value.

Saving A Filter

The filtered contents of a table can be saved as a query by selecting *Advanced|Save As Query*. Enter a name for the query and click *OK.* The query is now saved within the database.

Remove a Filter 

To view all records in a table again, click the depressed *toggle Filter* button.

Creating a Relationship

A primary feature of relational databases is that they contain multiple tables, each of which can have a relationship with any of the other tables.

Our database needs more than one table anyway because we need to be able to track not only customers but also products, as well as the products the customers actually purchase. So we need to create two new tables.

- *Products (ProductId, ProductName, Price, DateCreated)*
- *Orders (OrderId, CustomerId, ProductId, DateCreated)*

Now it's time to create the relationship between all three tables.

1. While viewing a table in Design view, DESIGN tab is selected, click Relationships from the Ribbon. A Show Table dialog box will pop up, displaying all three tables.
2. Select all of them and click Add.
3. You will now see three boxes that represent your three tables. Click and drag the CustomerId from the Customers table across to the corresponding CustomerId field on the Orders table.
4. The Edit Relationships dialog will pop up. Click Enforce Referential Integrity so that it is checked. Click OK. You will now see a line established between the CustomerId field on the Customers table and the CustomerId on the Orders table.

Relationship Types

We just established a many-to-many relationship. There are three types of relationships that you can establish between tables. These are as follows:

Many-To-Many Relationship

This is what our example above uses. A row in table A can have many matching rows in table B, and vice versa. In our case, a single customer can order many products, and a single product could have many customers.

One-To-Many Relationship

This is the most common relationship type. In this type of relationship, a row in table A can have many matching rows in table B, but a row in table B can have only one matching row in table A. For example, a row in a Gender table (which contains the records Male and Female) can have many matching rows in a Customers table, but a row in the Customers table can only have one matching row in the Gender table.

One-To-One Relationship

A row in table A can have only one matching row in table B, and vice versa. This is not a common relationship type, as the data in table B could just have easily been in table A. This relationship type is generally only used for security purposes, or to divide a large table, and perhaps a few other reasons.

Creating the Query

Now let's create a query that returns the names of all customers who have ordered a product.

1. Ensuring you have the CREATE tab open on the Ribbon, click Query Design.
2. The Show Table dialog box will appear with all of our tables listed. Select all three tables and click Add, then click Close.
3. The three tables are now represented in the top pane. Choose the fields you'd like to be presented in the results of your query. You can either double-click on the field name or click and drag it down to a column in the bottom pane. Select the fields.
4. Click the Run button at the top-left part of the Ribbon. You should now see the result of the query.
5. Save the query by right-clicking on the Query1 tab and giving it a name. Call it Customer Orders

Switch to SQL View

To switch to SQL view, you simply click the SQL icon at the bottom-right corner of Access. You can also choose SQL View from the View icon at the top-left of the Ribbon.

SQL Statements

In SQL view, you will see a SQL statement. This SQL statement represents the query that you constructed. If you're not familiar with SQL, this might look a little scary. But it needn't be that way. That SQL statement is simply extracting records from the database using our precise criteria.

SELECT The SELECT statement is used to select data from a database table.

- SELECT Customers.FirstName, Customers.LastName
  FROM Customers

WHERE The WHERE clause in SELECT statements can be used to filter the records.

- SELECT Customers.FirstName, Customers.LastName
  FROM Customers
  WHERE Customers.LastName = 'Boyle'

AND and OR operators AND and OR operators are used to make logical comparisions.

- SELECT Customers.FirstName, Customers.LastName
  FROM Customers
  WHERE Customers.LastName = 'Boyle' AND Customers.CustomerId = 1;
- SELECT Customers.FirstName, Customers.LastName
  FROM Customers
  WHERE Customers.LastName = 'Boyle' OR Customers.CustomerId = 1;

ORDER BY The ORDER BY clause in SELECT statement is used to sort the result set in ascending or descending order. The ORDER BY clause sorts the records in ascending order by default.

- SELECT Customers.FirstName, Customers.LastName
  FROM Customers
  ORDER BY Customers.FirstName ASC/DESC;

AGGREGATE FUNCTIONS

An aggregate function performs a calculation on a set of values and returns a single value. Some common aggregate functions are given below.

SUM

It calculates the sum for a set of values.

- SELECT SUM(Products.Price) FROM Products;

MAX

Returns the maximum value from a set of values.

- SELECT MAX(Products.Price) FROM Products;

MIN

Returns the minimum value from a set of values.

- SELECT MIN(Products.Price) FROM Products;

COUNT

Counts the number of rows in a table.

- SELECT COUNT(Products.ProductId) FROM Products;

As you can see, each time we add criteria to our query in Design view, that criteria is added to the SQL statement, and we can limit the results by only those exact records that we are interested in.

Switch Back to Datasheet View

Switch Back to Datasheet View to see the result.

Lab Session 8

OBJECTIVE(S)

- Learn about Relational Database Management Systems
- Learn about Creating Database in Microsoft Access
- Learn about Creating Relationships
- Learn about Creating Queries

Lab Tasks

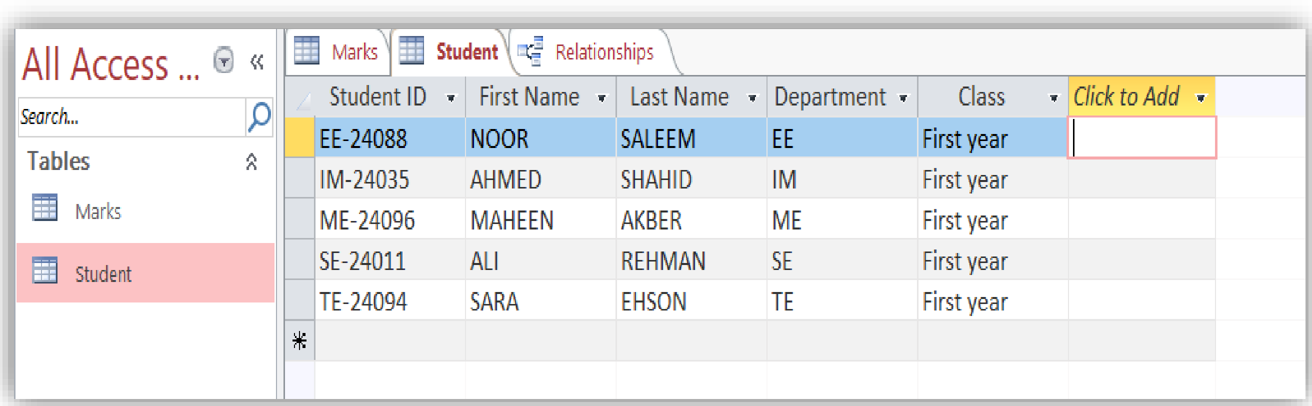
Create the following tables in Microsoft Access:

Table1: *Student* (containing fields: *Student ID, First Name, Last Name, Department* and *Class).*

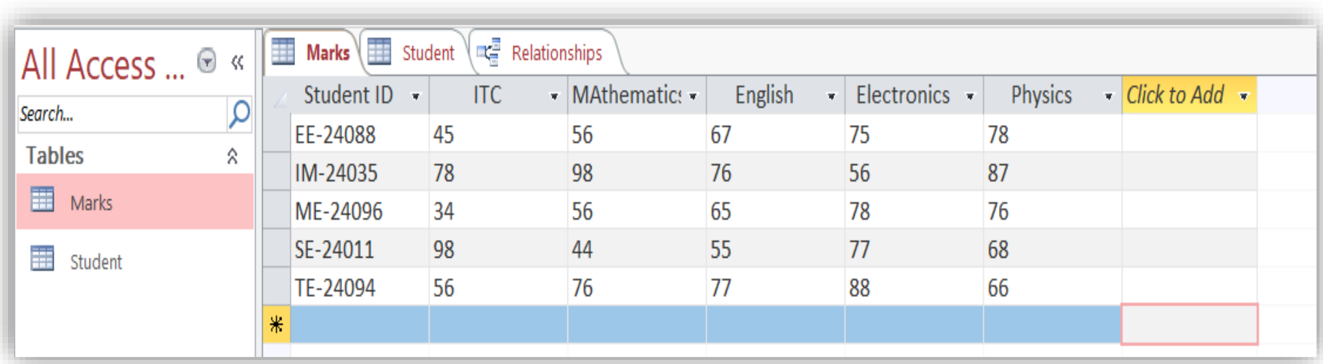
Table2: *Marks* (containing fields: *Student ID, ITC, Mathematics, English, Electronics, Physics).*

Select appropriate primary keys and define appropriate relationship for these two tables. Add some sample data to the tables and make a hard copy of the tables created.

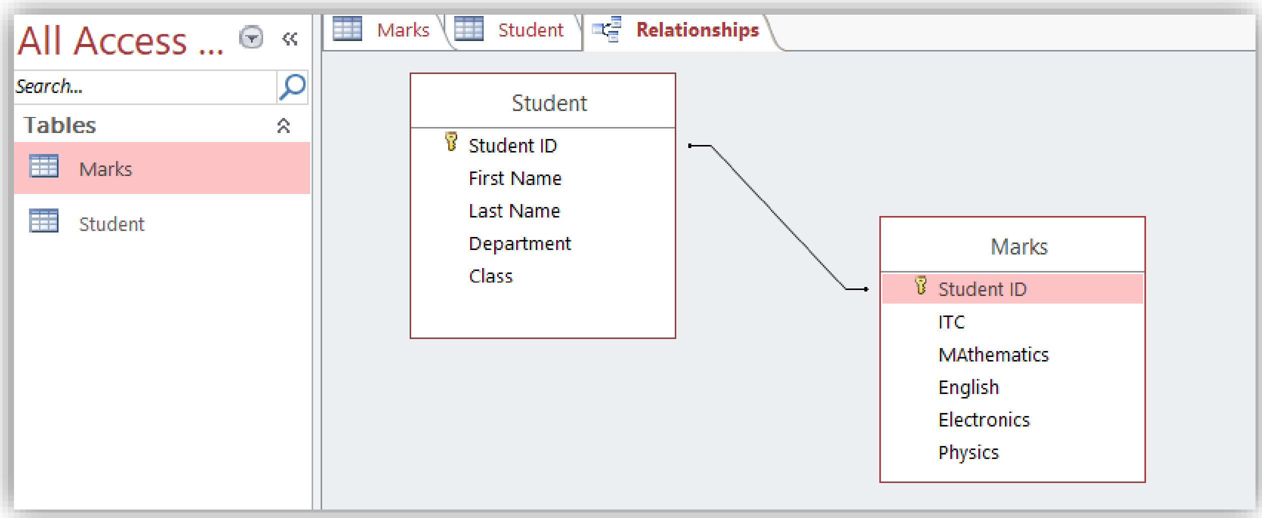
Attach your printout here.



Student ID	First Name	Last Name	Department	Class	Click to Add
EE-24088	NOOR	SALEEM	EE	First year	
IM-24035	AHMED	SHAHID	IM	First year	
ME-24096	MAHEEN	AKBER	ME	First year	
SE-24011	ALI	REHMAN	SE	First year	
TE-24094	SARA	EHSON	TE	First year	



Student ID	ITC	MAthematic:	English	Electronics	Physics	Click to Add
EE-24088	45	56	67	75	78	
IM-24035	78	98	76	56	87	
ME-24096	34	56	65	78	76	
SE-24011	98	44	55	77	68	
TE-24094	56	76	77	88	66	



Explanation of the Relationships

- **Student ID** serves as a **Primary Key** in the **Student** table and as a **Foreign Key** in the **Marks** table.
- This setup establishes a **one-to-many** relationship, meaning each student can have multiple records in the **Marks** table, but each mark record is associated with only one student.

Lab Session 9

OBJECT

Creating queries, forms and reports in Microsoft Access

INTRODUCTION TO QUERIES

Queries select records from one or more tables in a database so they can be viewed, analyzed, and sorted on a common datasheet. The resulting collection of records, called a dynaset (short for dynamic subset), is saved as a database object and can therefore be easily used in the future. The query will be updated whenever the original tables are updated. Types of queries are *select queries* that extract data from tables based on specified values, *find duplicate* queries that display records with duplicate values for one or more of the specified fields, and *find unmatched* queries display records from one table that do not have corresponding values in a second table.

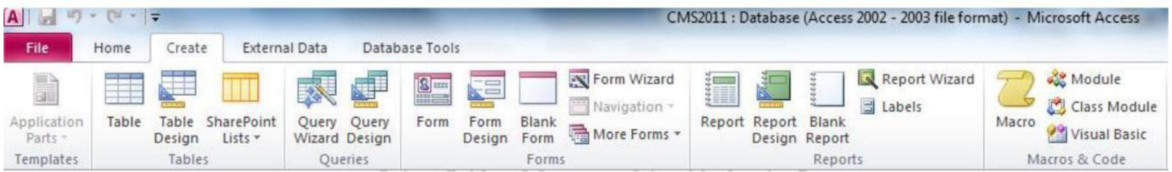


Figure 12.1 Create Menu Quick Access Toolbar

Create a Query in Design View

From the Create Main Menu as in figure 12.1, click the *Query Design icon*. Select tables and existing queries from the *Tables* and *Queries* tabs and click the *Add* button to add each one to the new query. Click *Close* when all of the tables and queries have been selected. Add fields from the tables to the new query by double-clicking the field name in the table boxes or selecting the field from the *Field:* and *Table:* drop-down menus on the query form. Specify sort orders if necessary.

Enter the criteria for the query in the *Criteria:* field. The *Expression Builder*  can also be used to assist in writing the expressions. After you have selected all of the fields and tables, click the *Run* button on the toolbar. Save the query by clicking the *Save* button.

Query Wizard

Click the *Create query by using wizard* icon in the database window to have Access step you through the process of creating a query. From the first window, select fields that will be included in the query by first selecting the table from the drop-down *Tables/Queries* menu. Select the fields by clicking the **>** button to move the field from the Available Fields list to Selected Fields. Click the double arrow button >> to move all of the fields to Selected Fields. Select another table or query to choose from more fields and repeat the process of moving them to the Selected Fields box. Click *Next* > when all of the fields have been selected. On the next window, enter the name for the query and click *Finish.*

Find Duplicates Query

This query will filter out records in a single table that contain duplicate values in a field. Click the *New* button on the Queries database window, select *Find Duplicates Query Wizard* from the *New Query* window and click *OK*. Select the table or query that the find duplicates query will be applied to from the list provided and click *Next* >. Select the fields that may contain duplicate values by highlighting the names in the Available fields list and clicking the **>** button to individually move the fields to the Duplicate-value fields list or **>>** to move all of the fields. Click *Next* > when all fields have been selected. Select the fields that should appear in the new query along with the fields selected on the previous screen and click *Next* >. Name the new query and click *Finish*.

Delete a Query

To delete a query, select the required query and press the *Delete* key on the keyboard.

FORMS

Forms are used as an alternative way to enter data into a database table.

Create Form by Using Wizard

From the Create Main Menu as in figure 12.1, click *Form wizard* option in Forms section. From the *Tables/Queries* drop-down menu, select the table or query whose datasheet the form will modify. Then, select the fields that will be included on the form by highlighting each one the *Available Fields* window and clicking the single right arrow button **>** to move the field to the *Selected Fields* window. To move all of the fields to Select Fields, click the double right arrow button **>>**. If you make a mistake and would like to remove a field or all of the fields from the Selected Fields window, click the left arrow **<** or left double arrow << buttons. After the proper fields have been selected, click the *Next* > button to move on to the next screen. On the second screen, select the layout of the form.

- **Columnar** - A single record is displayed at one time with labels and form fields listed side-by-side in columns.
- **Justified** - A single record is displayed with labels and form fields are listed across the screen.
- **Tabular** - Multiple records are listed on the page at a time with fields in columns and records in rows.
- **Datasheet** - Multiple records are displayed in Datasheet View

Click the ***Next*** **>** button to move on to the next screen. Select a visual style for the form from the next set of options and click *Next* >. On the final screen, name the form in the space provided. Select "Open the form to view or enter information" to open the form in Form View or "Modify the form's design" to open it in Design View. Click *Finish* to create the form.

Adding Records Using A Form

Input data into the table by filling out the fields of the form. Press the *Tab* key to move from field to field and create a new record by clicking *Tab* after the last field of the last record. A new record can also be created at any time by clicking the *New Record* button at the bottom of the form window. Records are automatically saved as they are entered so no additional manual saving needs to be executed.

Editing Forms

The following points may be helpful when modifying forms in Design View:

Grid lines - By default, a series of lines and dots underlay the form in Design View so form elements can be easily aligned. To toggle this feature on and off right click and select Grid.
Resizing Objects - Form objects can be resized by clicking and dragging the handles on the edges and corners of the element with the mouse.
Change form object type - To easily change the type of form object without having to create a new one, right click on the object with the mouse and select *Change To* and select an available object type from the list.
Label/object alignment - Each form object and its corresponding label are bounded and will move together when either one is moved with the mouse. However, to change the position of the object and label in relation to each other (to move the label closer to a text box, for example), click and drag the large handle at the top, left corner of the object or label.
Tab order - Alter the tab order of the objects on the form by selecting *Tab Order* from design Menu when Form is active. Click the gray box before the row you would like to change in the tab order, drag it to a new location, and release the mouse button.
Form Appearance - Change the background color of the form by clicking the *Shape Fill* button on the format menu inside Form Design. Change the color of individual form objects by highlighting one and selecting a color from

Color palette on the formatting toolbar. The font and size, font effect, font alignment, border around each object, the border width, and a special effect can also be modified using the formatting toolbar buttons inside Form Design Tools.

Form Controls

This section explains the uses for other types of form controls including lists, combo boxes, checkboxes, option groups, and command buttons.

- ***List and Combo Boxes:*** If there are small, finite number of values for a certain field on a form, using combo or list boxes may be a quicker and easier way of entering data. These two control types differ in the number of values they display. List values are all displayed while the combo box values are not displayed until the arrow button is clicked to open.

- ***Check Boxes and Option Buttons:*** Use check boxes and option buttons to display yes/no, true/false, or on/off values. Only one value from a group of option buttons can be selected while any or all values from a check box group can be chosen. Typically, these controls should be used when five or fewer options are available. Combo boxes or lists should be used for long lists of options.

- ***Command Buttons:*** Command buttons provide you with a way of performing action(s) by simply clicking them. When you choose the button, it not only carries out the appropriate action, it also looks as if it's being pushed in and released.

Reports

Reports will organize and group the information in a table or query and provide a way to print the data in a database.

Using the Wizard

From the Create Main Menu as in figure 12.1, click Report Wizard option in the Reports section. Select the information source for the report by selecting a table or query from the *Tables/Queries* drop-down menu. Then, select the fields that should be displayed in the report by transferring them from the *Available Fields* menu to the *Selected Fields* window using the single right arrow button **>** to move fields one at a time or the double arrow button **>>** to move all of the fields at once. Click the *Next* > button to move to the next screen.

Select fields from the list that the records should be grouped by and click the right arrow button **>** to add those fields to the diagram. Use the *Priority* buttons to change the order of the grouped fields if more than one field is selected. Click *Next* > to continue. If the records should be sorted, identify a sort order here. Select the first field that records should be sorted by and click the A-Z sort button to choose from ascending or descending order. Click *Next* > to continue. Select a layout and page orientation for the report and click *Next*
Select a color and graphics style for the report and click *Next* >. On the final screen, name the report and select to open it in either Print Preview or Design View mode. Click the *Finish* button to create the report.

Create in Design View

Click Report Design button in the Reports section of create access toolbar and you will be presented with a blank grid. Automatically properties window will open for this newly being created report. Click on Record Source property combo box and you will be presented with the list of available tables and queries. Choose the data source of the report from the drop-down menu and click *OK*. Design the report in much the same way you would create a form. For example, double-click the title bar of the Field Box to add all of the fields to the report at once. Then, use the handles on the elements to resize or move them to different locations, and modify the look of the report as per your requirements.

Lab Session 9

OBJECT

Creating queries, forms and reports in Microsoft Access

LAB TASK:

4. Create the following tables in a Database named *Job Details* and add at least 5 records.
   - *Employees(employeeID, employeeName, hiringDate, deptID, jobID)*

Employees

employee ID	employee Na	hiring date	dept ID	job ID	Click to Add
123	Fahad	27/11/2022	6	20	
145	Lisa	08/06/2022	3	77	
456	Hareem	22/08/2008	1	3	
789	Isha	03/05/2005	4	56	
980	Haley	13/01/2023	4	67	

   - *Departments(deptID, deptName, locationID)*

Employees | Departments

Dept ID	Dept Name	Location ID	Click to Add
1	HR	g3	
2	Finance	f2	
3	Marketing	h4	
4	Sales	e1	

   - *Jobs(jobID, jobTitle, salary)*

Employees | Departments | Jobs

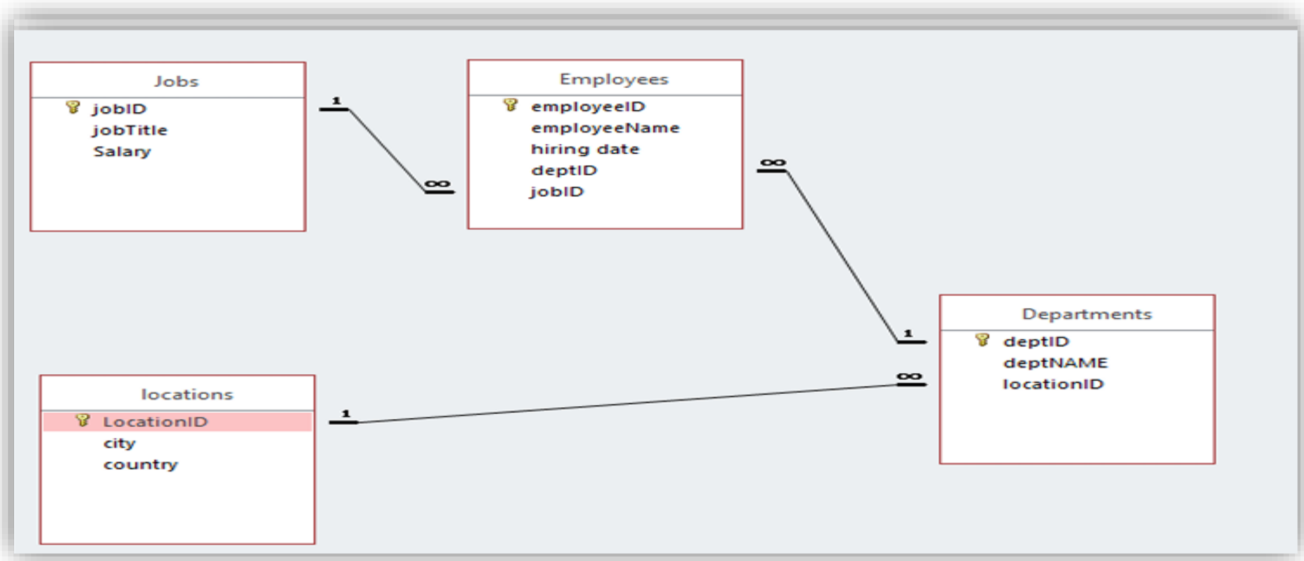
job ID	job title	salary	Click to Add
1	Manager	Rs20,000.00	
2	Analyst	Rs15,000.00	
3	Developer	Rs10,000.00	
4	Technician	Rs5,000.00	
		Rs0.00	

   - *Locations(locationID, city, country)*

Employees | Departments | Jobs | Locations

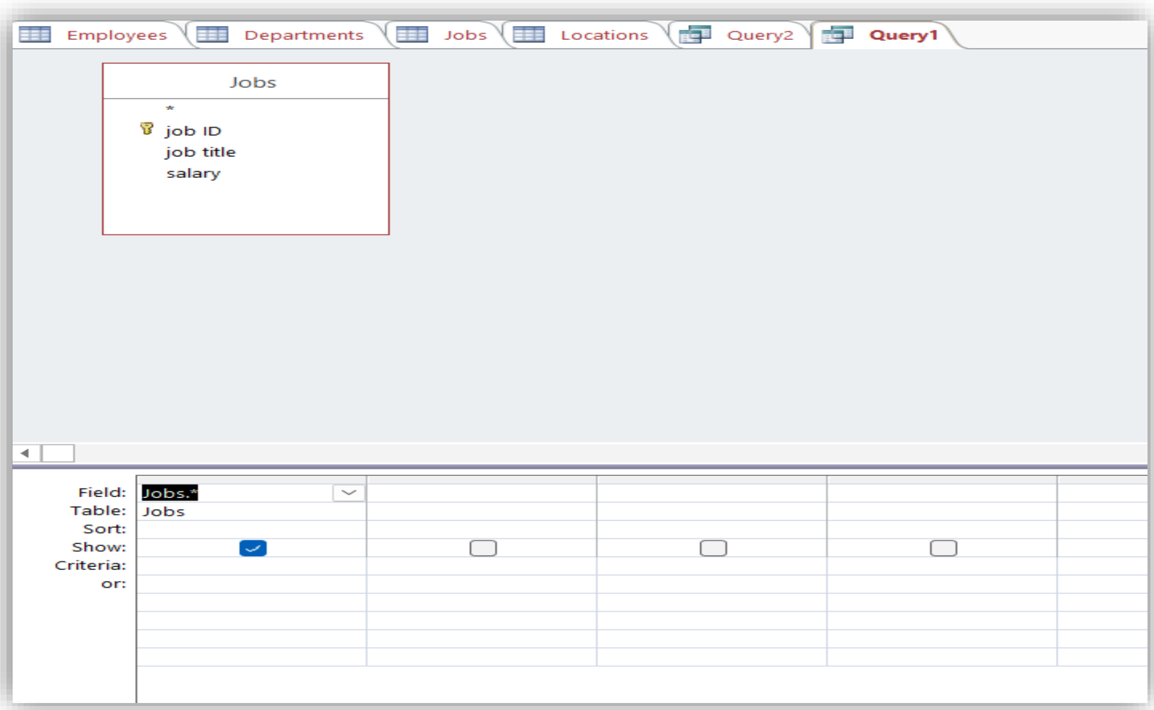
location ID	city	country	Click to Add
e1	Karachi	Pakistan	
f2	Lahore	Pakistan	
g3	New York	USA	
h4	Toronto	Canada	

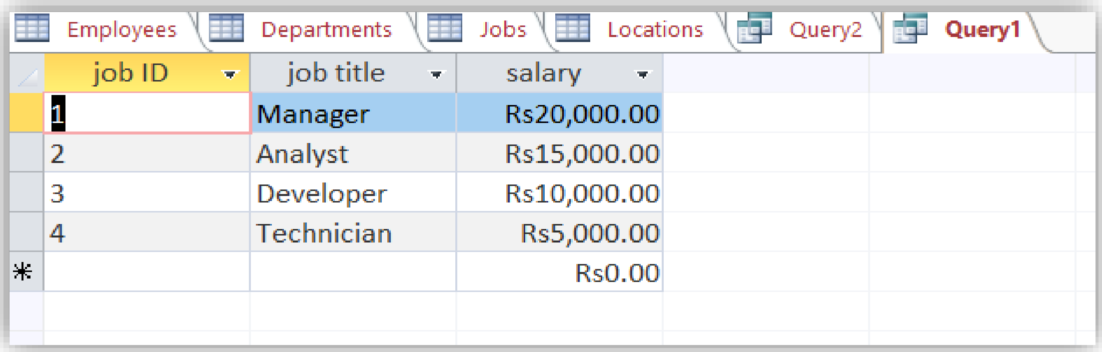
5. Create appropriate relations between the newly created tables. Ensure referential integrity.



6. Write suitable queries for each of the following statements.

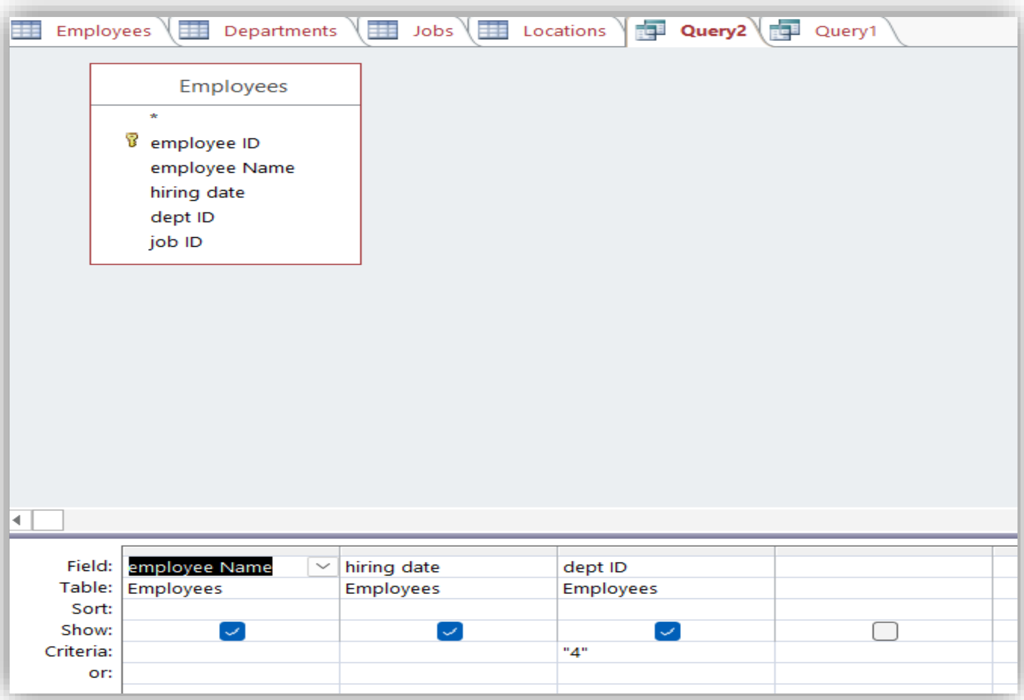
(i) Display all the records from the Jobs table.

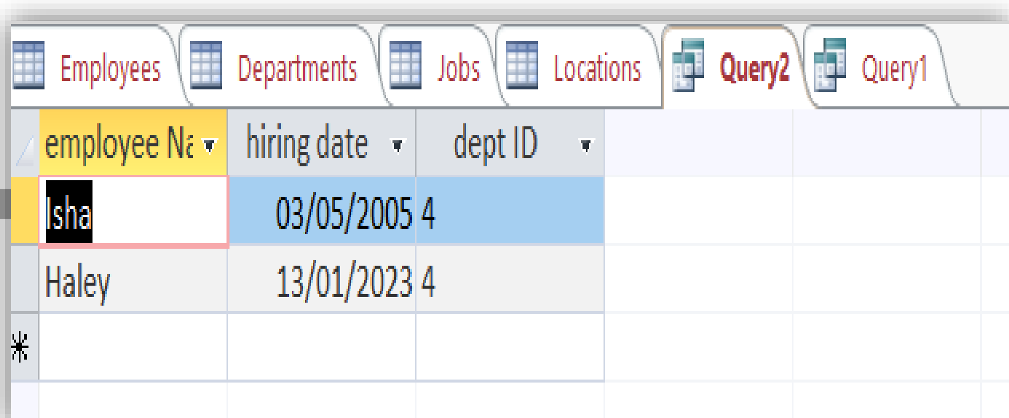




job ID	job title	salary
1	Manager	Rs20,000.00
2	Analyst	Rs15,000.00
3	Developer	Rs10,000.00
4	Technician	Rs5,000.00
		Rs0.00

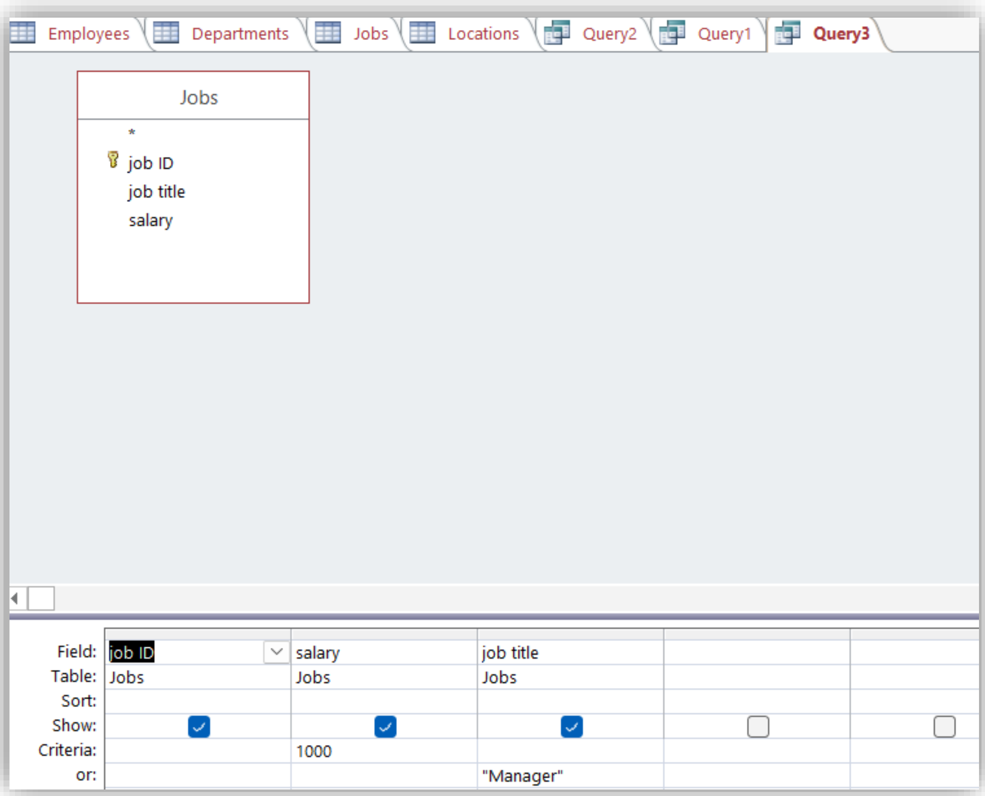
(ii) Display the hiring date of the employees with department ID = “4”.

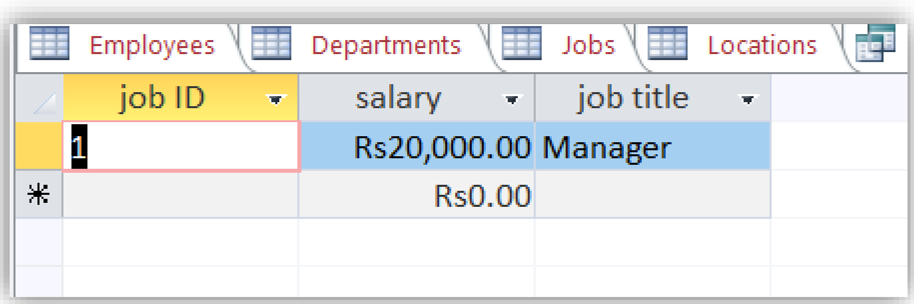




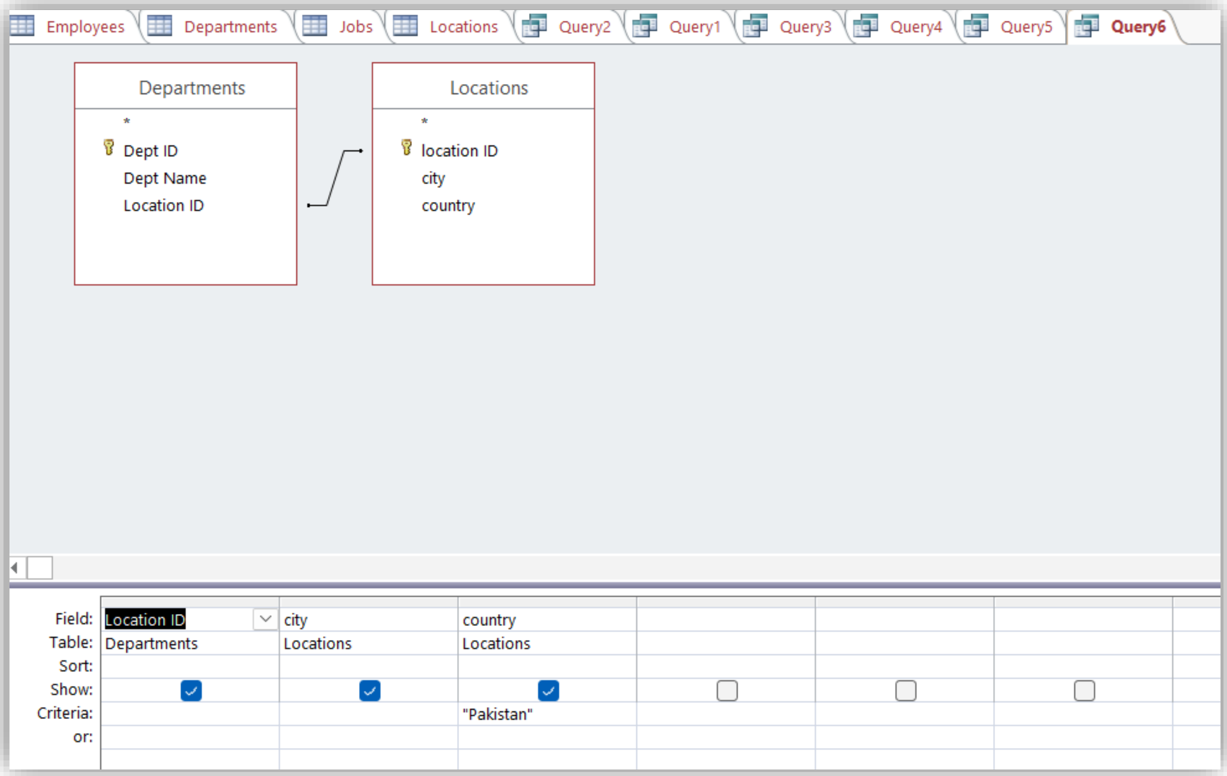
employee Na	hiring date	dept ID
Isha	03/05/2005	4
Haley	13/01/2023	4

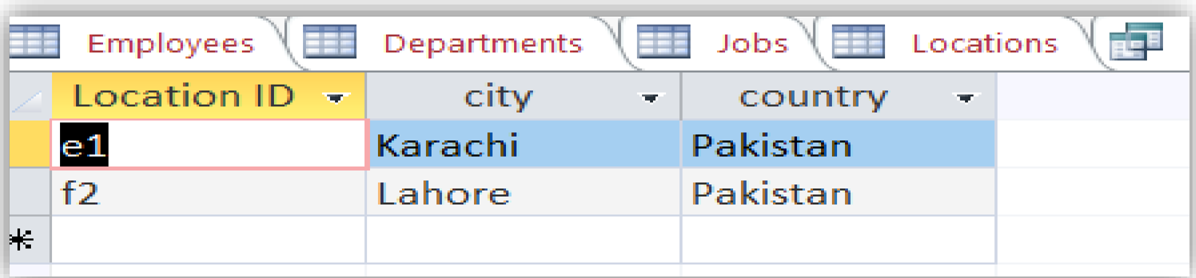
(iii) Display job IDs for which the salary is greater than “$1000” or Title is “Manager”.



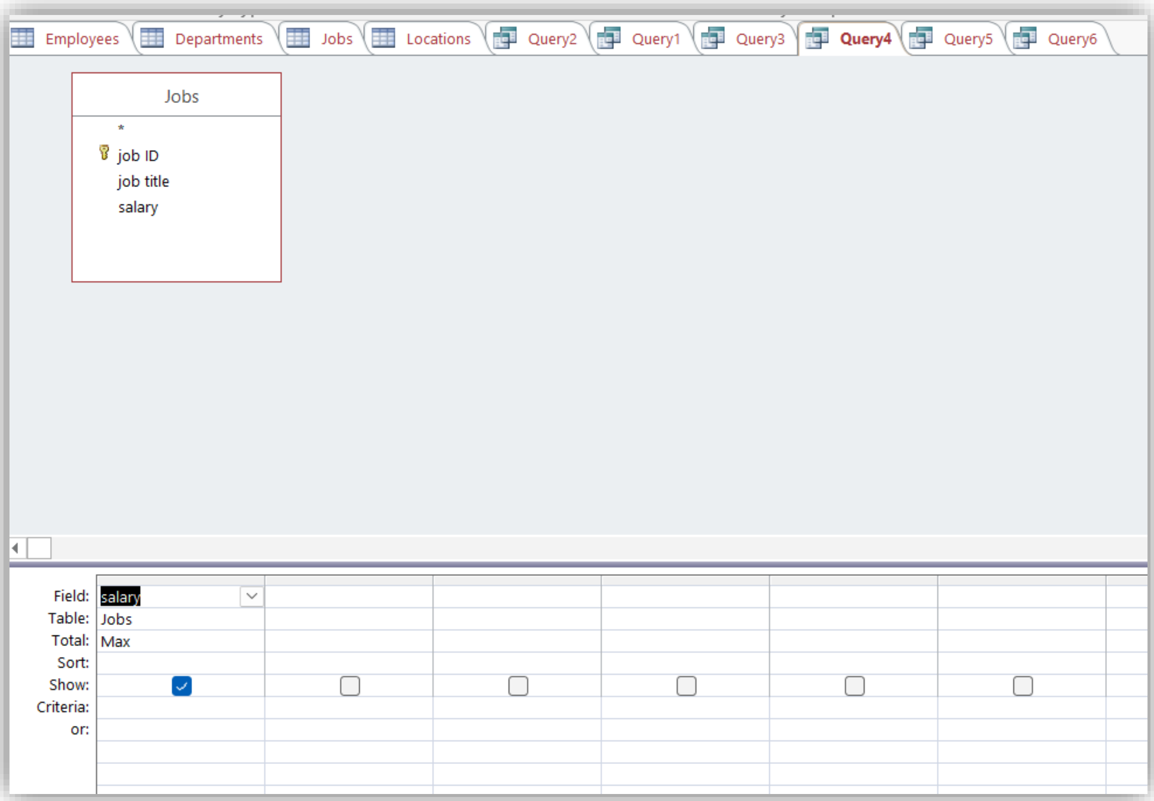


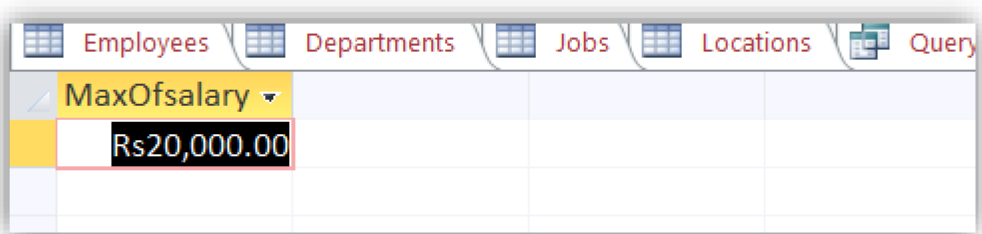
(iv) Display the location IDs and cities of the departments that operate in Country of Pakistan.



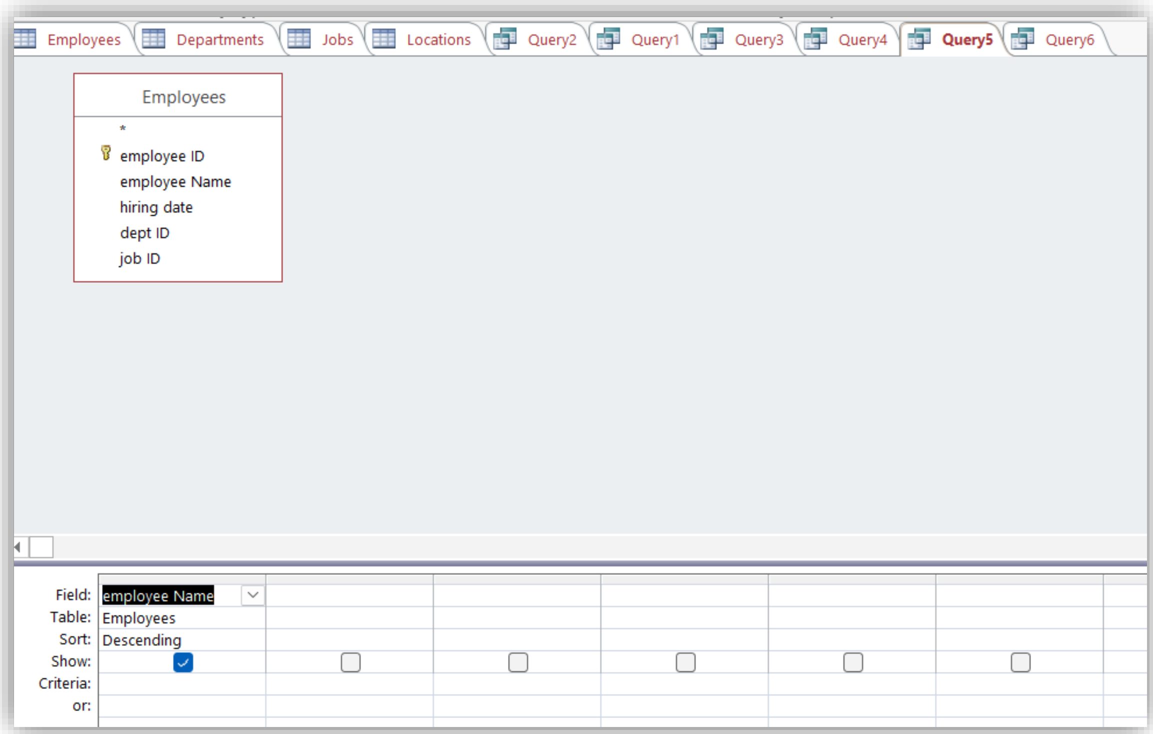


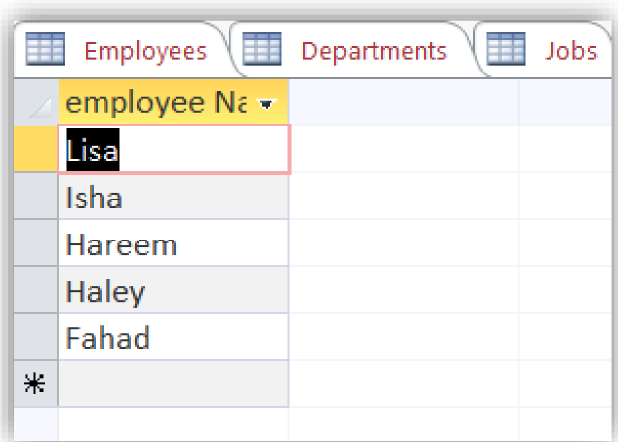
(v) Display the highest paid salary.





(vi) Display the names of the employees in reverse alphabetical order.





SUBMISSION GUIDELINES

- Take a screenshot of each task.
- Place all the screenshots in a single word file labeled with Roll No and Lab No. e.g. **'SE202xxx_Lab09'.**
- Convert the file into PDF.
- Place all the related files along with the PDF file in a folder labeled with Roll No and Lab No. e.g. 'SE 202xxx_Lab09'.
- Submit the folder at GCR
- **-100%** policies for plagiarism.

NED University of Engineering & Technology
Department of Software Engineering

Course Code & Title: CT-174 FIT ____ Assessment Rubric for Psychomotor (P2-PLO-3)

Instructor: _________ Batch: 2024 Class: FESE Semester: Fall 2024.
Student's Name:_____ Roll No.: _______________
Laboratory Session No. _________ Date:

Psychomotor Domain Assessment Rubric-Level P2-PLO3					
Skill Sets	Extent of Achievement				
	Below Average (0--1)	**Average (2)**	**Good (3)**	**Very Good (4)**	**Excellent (5)**
Problem Definition and Understanding	Unable to define and understand the problem clearly.	Struggles to define the problem clearly, with significant gaps in understanding	Defines the problem, but with gaps or a limited understanding of key components, impacting the overall quality of definition.	Effectively defines the problem with a good understanding of its components, showcasing proficiency in problem definition.	Demonstrates an exceptional understanding of the problem, identifying all key components and challenges with insightful depth.
Technical Implementation ***Proficient use of tools to complete the problem***	Struggles to use tools effectively, requiring significant improvement.	Uses tools with noticeable inefficiencies.	Demonstrates basic proficiency in utilizing tool, with some room for improvement.	Effectively uses tools to support the task.	Integrates use of tools seamlessly to enhance the design and development process
Solution Design& Accuracy	Struggles to relate the implemented solutions to the design and development goals, requiring significant improvement.	Shows limited connection between the implemented solutions and the design and development context.	Demonstrates a basic understanding of how the solutions align with design and development requirements.	Effectively relates the implemented solutions to the overall design and development objectives	Maps the implemented solutions seamlessly to the overarching design and development goals, demonstrating a comprehensive understanding.
Trouble Shooting Skills Issue resolution skills	Unable to trouble shoot	Limited troubleshooting	Adequate troubleshooting, minor errors remaining	Good troubleshooting, no errors remaining	Excellent trouble shooting no errors remain without delay.
Document Organization and on time submission	Poor Unclear disorganized document late submission	Somewhat clear with lack of organization and structuring late submission	Somewhat clear with lack of organization and structuring on time submission	Reasonably clear and organized with adequate structuring on time submission	Highly clear and well-structured document, On time submit

Weighted CLO (Psychomotor Score)		
Instructor's Signature with Date		

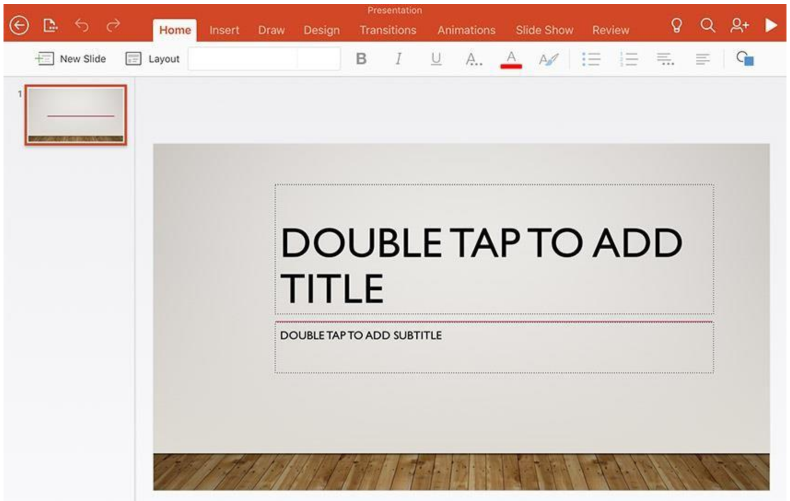
LAB 10 – MS PowerPoint

OBJECTIVE(S)

- Learn how to use Microsoft PowerPoint
- Create a presentation
- Use drawing tools in PowerPoint
- Add animation to your presentation

Introducing Microsoft PowerPoint

Microsoft PowerPoint is an application used to create presentations. Instead of the older method of using overhead slides and projectors, PowerPoint is a flexible, fast way to create professional-looking visuals for many different types of presentations. If you haven't already used PowerPoint for school projects, you will very likely need to use it soon.



Slide Layout and Style

Start with a blank template. The first slide will be your title slide. You can pick any layout you want. Just click in the boxes to add text. To add a new slide, go to Insert → New Slide. Again you will be asked to select a layout. Pick anything you want. If you ever want to change the layout of your slide, go to Format → Slide Layout. You can now select a new slide layout.
In addition to layout, you can set the style of your slides. Go to Format → Apply Design Template or Format → Slide Layout (it depends on what version of PowerPoint you're using). By clicking on the different styles in the left list, you should see a preview of the style on the right. Select whatever style you like. You can always change it later.

MS PowerPoint PPT Templates

PowerPoint Templates (frequently also called "themes") are the quickest way to add style to your Microsoft PowerPoint presentation and one of the reasons that PowerPoint is so popular.

The easiest way to add a design to your slide is to use a template, which you'll find on PowerPoint's ribbon on the **Design** tab. Click on one of the template thumbnails to apply it to your presentation.

Inserting a Picture

From within PowerPoint: To insert an image from the files that PowerPoint already has, go to Insert → Picture. You can then select "Clip Art", which is provided by Microsoft. Other options for graphics and images inside of PowerPoint are also on the Insert menu. Some examples are Word Art, Autoshapes, Organization Chart, and Word Table.

From the Internet: Many images, graphs, and charts on the Internet are available for use. First they must be saved onto your computer. For Windows, you do that by left-clicking on the image, selecting "Save Picture As", and then putting the picture onto one of your disks. For Macs, you hold down the apple button, click, and then choose the "Download Image to Disk" option. Once the graphic is on your computer, go to Insert → Picture → From File and then select the image.

Animating the Presentation

You can animate your text, inserted objects (like pictures and graphs) or slide transition. We'll show you a few basics.

Animating Text

The most common animation effect for text is to have each line or bullet appoint appear one at a time. Select a slide with bulleted text. Go to Tools → Build Slide Text. This will present you with several animation choices. First try "Fly from Left". Now let's see what it looks like. Go to View → Slide Show. Each time you hit the space bar or click with your mouse, another bullet point will fly in. A more subtle and probably more professional choice is Dissolve. Go to Tools → Build Slide Text → Dissolve. View the slide show again.

Select a slide with bulleted text. Go to Slide Show → Animation Schemes. The Animation Schemes task pane will appear on the right side of your PowerPoint window. First try "Unfold" in the Moderate section. A preview of the animation shows immediately, and you can "run" the slide show again by going to Slide Show → View Show. If you don't see it on the menu right away, hover your mouse over the menu for a moment and it will appear.

Animating Pictures

This will be similar to animating text. Find a slide with pictures or graphs on it. Click on the picture/graph to select it. Go to Tools → Animation Settings. A pop up window should appear. If the Build Option is set to No Build, change it to Build. Under Effects, there will be a pulldown option with different animations for how you want the picture to appear. You can test the different animations styles by going to Slide Show view.

Slide Transition

You can also animate the way the next slide appears. Go to Tools → Slide Transition for your options. Go to Slide Show view to see what it looks like.

LAB 10 – MS PowerPoint

OBJECTIVE(S)

- Learn how to use Microsoft PowerPoint
- Create a presentation
- Use drawing tools in PowerPoint
- Add animation to your presentation

EXERCISES

- Create a Power Point presentation on a topic of your own choice of 2 slides. Add appropriate animation effects and clip arts, smart art, graphs, photos videos, etc to make a powerful presentation.

Attach the colored printout here.

NED University of Engineering & Technology
Department of Software Engineering

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